

PART A

1. What are statically and dynamically induced emfs? Explain.

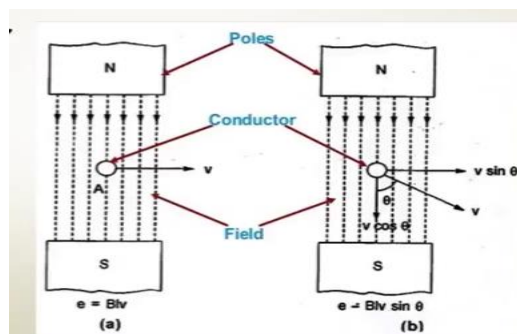
Statically induced emf - conductor remains stationary and magnetic field linked with it is changed

Dynamically induced emf - conductor is moving and magnetic field remains stationary

Dynamically induced emf

It is produced when a conductor is moved across a stationary field

Eg. Generation of emf in electric generators



B = flux density in Wb/m^2

l = active length of conductor in metres.

v = velocity in m/sec.

∴

$$e = B l v \quad \text{volts}$$

This is the induced e.m.f. when plane of motion is exactly perpendicular to the plane of flux. This is maximum possible e.m.f. as plane of motion is at right angles to plane of the flux.

If the direction of movement is at an angle θ from direction of flux, as shown in fig b, then induced emf is

$$e = B l v \sin \theta \quad \text{volts}$$

Statically induced emf

Conductor- stationary

Magnetic field moving or changing

Statically induced emf are of two types

1. Self induced emf
2. Mutually induced emf

Self induced emf

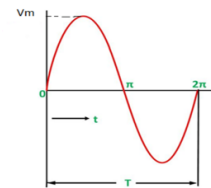
Emf induced in a coil due to change of its own flux linked with it is called self induced emf, ie emf is induced due to the change of current in the same coil itself.

Mutually induced emf

The emf induced in the coil due to change in current in neighboring coil is called mutually induced emf

2. Derive the expression for average value of a sinusoidal wave form.

(i) $v = V_m \sin \theta$



$$\begin{aligned} V_{avg} &= \frac{1}{T} \int_0^T v dt \\ &= \frac{1}{T} \int_0^T v dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} V_m \sin \theta d\theta \\ &= \frac{-V_m}{2\pi} [\cos \theta]_0^{2\pi} \\ &= \frac{-V_m}{2\pi} [\cos 2\pi - \cos 0] \\ &= \frac{-V_m}{2\pi} (1 - 1) \\ &= \frac{-V_m}{2\pi} (0) = 0 \end{aligned}$$

3. Derive an expression for the energy stored in an inductor.

- An inductor is a passive electronic component that stores energy in the form of a magnetic field.
- Unit of inductance is henry(H)
- The relation between voltage and current in an inductor is written as

$$v = L \frac{di}{dt}$$

Where v is the voltage across inductor in volts, and 'i' is the current through inductor in amps. We can rewrite the above equations as

$$di = \frac{1}{L} v dt$$

- Integrating on both sides, we get

$$\int_0^t di = \frac{1}{L} \int_0^t v dt$$

$$i(t) - i(0) = \frac{1}{L} \int_0^t v dt$$

$$i(t) = \frac{1}{L} \int_0^t v dt + i(0)$$

The power absorbed by inductor is

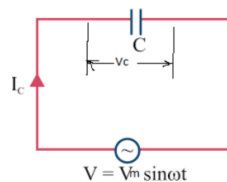
$$P = vi = Li \frac{di}{dt} \text{ watts}$$

The energy stored by the inductor is

$$E = \int_0^t p dt = \int_0^t Li \frac{di}{dt} dt = \frac{Li^2}{2}$$

$$E = \frac{1}{2} Li^2$$

- 4. Prove that in a purely capacitive circuit the current leads the applied voltage by 90 degrees and the power consumed is zero.**



- Consider an ac circuit consisting of a pure capacitance (C Farads) shown in figure.
- Let the applied voltage given by the equation be $v = V_m \sin \omega t$
- Capacitor is the element that can store charge in the electric field

$$q \propto v$$

$$Q = Cv$$

$$i = \frac{dq}{dt}$$

$$i = \frac{d(Cv)}{dt}$$

$$i = C \frac{dv}{dt}$$

$$i = C \frac{d(V_m \sin \omega t)}{dt}$$

$$i = C V_m \cos \omega t * \omega$$

$$i = \omega C V_m \cos \omega t$$

$$i = \omega C V_m \sin(\omega t + \frac{\pi}{2})$$

$$i = \frac{V_m}{\frac{1}{\omega C}} \sin(\omega t + \frac{\pi}{2})$$

$$\text{Where } I_m = \frac{V_m}{\frac{1}{\omega C}} = \frac{V_m}{X_C}$$

Here the term $\frac{1}{\omega C}$ determines the magnitude of the current through the capacitor in relation to the applied voltage. This term is called capacitive reactance, X_C

$$\text{capacitive reactance, } X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

- Now the expression for current can be written as $i = I_m \sin(\omega t + \frac{\pi}{2})$

$$\text{Where } I_m = \frac{V_m}{X_C}$$

For a purely capacitive circuit, current leads the applied voltage by 90°

Average power consumed by capacitor over full cycle

$$P_{avg} = \frac{1}{T} \int_0^T P_i dt$$

$$= \frac{1}{2\pi} \int_0^{2\pi} v * i dt$$

$$= \frac{1}{2\pi} \int_0^{2\pi} V_m \sin \omega t * I_m \sin(\omega t + \frac{\pi}{2}) d\omega t$$

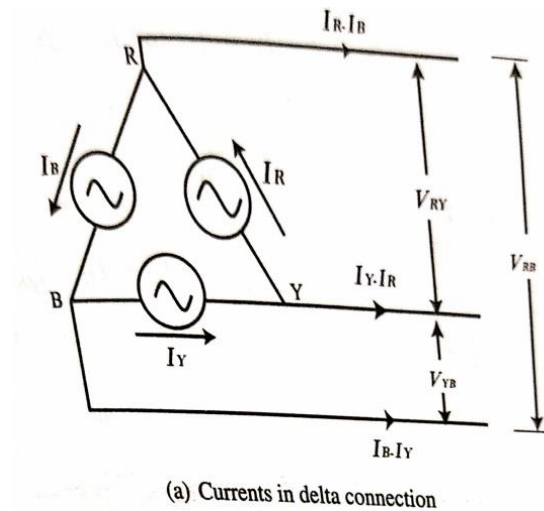
$$= \frac{V_m I_m}{2\pi} \int_0^{2\pi} \sin \omega t * \cos \omega t d\omega t$$

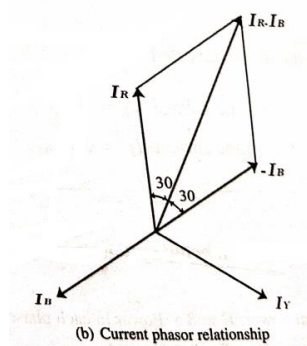
$$= \frac{V_m I_m}{4\pi} \int_0^{2\pi} 2 \sin \omega t * \cos \omega t d\omega t$$

$$\begin{aligned}
&= \frac{V_m I_m}{4\pi} \int_0^{2\pi} \sin 2\omega t d\omega t \\
&= \frac{V_m I_m}{4\pi} \left(\frac{-\cos 2\omega t}{2} \right) \Big|_0^{2\pi} \\
&= \frac{-V_m I_m}{4\pi} \left[\frac{\cos 4\pi - \cos 0}{2} \right] \\
&= \frac{-V_m I_m}{4\pi} [1 - 1] \\
P_{avg} &= 0
\end{aligned}$$

5. **Derive the relation between line and phase currents in a 3 phase delta connected system.**

- In Delta(Δ) interconnection, the dissimilar ends of the three phase windings are joined together i.e., finishing end of one phase is connected to the starting end of the other phase winding and so on as shown in figure.
- Here V_{RY} , V_{YB} and V_{BR} represents line voltages. Since only one phase winding is connected across two lines, The line voltage V_L is equal to phase voltage, V_{ph} .
- The phase relation between the phase currents I_R , I_Y and I_B is shown in figure.





- The line currents are $I_R - I_B$, $I_B - I_Y$ and $I_Y - I_R$,

One of the line currents $I_R - I_B$

$$= \sqrt{I_R^2 + I_B^2 + 2I_R I_B \cos 60}$$

$$= \sqrt{I_{ph}^2 + I_{ph}^2 + 2I_{ph}^2 \cos 60}$$

Where $I_R = I_B = I_{ph}$

$$= \sqrt{3I_{ph}^2} = \sqrt{3} I_{ph}$$

Hence for a Delta or Δ - connected system

Line voltage, $V_L = V_{ph}$

Line current, $I_L = \sqrt{3} I_{ph}$

- Relation between phase voltage and Line voltage in delta connected 3 – phase system is

$$V_L = V_{ph}$$

Now, $I_L = \sqrt{3} I_{ph}$

Similarly in star connection

$$V_L = \sqrt{3} V_{ph}$$

$$I_L = I_{ph}$$

- Equation of active power in 3 – phase system

$$P = \sqrt{3} V_{ph} I_{ph} \cos \phi$$

$$\text{Active power, } P = \sqrt{3} V_L I_L \cos \phi$$

Reactive Power

$$Q = \sqrt{3} V_{ph} I_{ph} \sin \phi$$

$$Q = \sqrt{3} V_L I_L \sin \phi$$

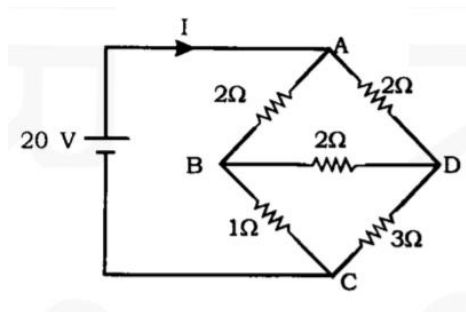
Apparent power

$$S = \sqrt{3} V_{ph} I_{ph}$$

$$S = \sqrt{3} V_L I_L$$

PART B

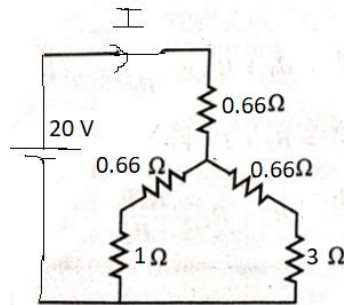
6. Find the source current I in the below figure using star-delta transformation.



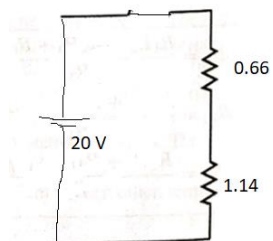
$$R_1 = \frac{2 * 2}{2 + 2 + 2} = 0.66$$

$$R_2 = \frac{2 * 2}{2 + 2 + 2} = 0.66$$

$$R_3 = \frac{2 * 2}{2 + 2 + 2} = 0.66$$



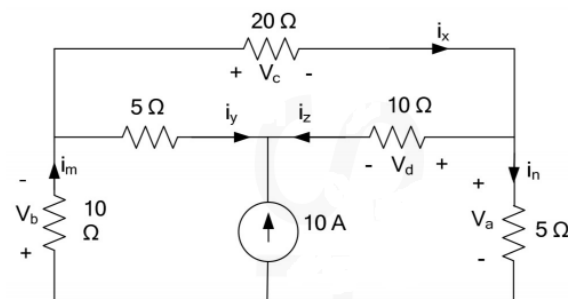
$$R_p = \frac{1.66 * 3.66}{1.66 + 3.66} = 1.14\Omega$$



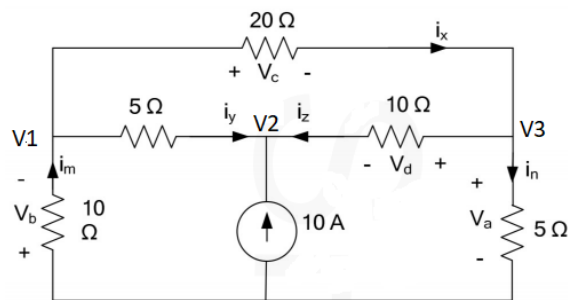
$$R_{eq} = 0.66 + 1.14 = 1.802 \, \Omega$$

$$I = \frac{V}{R} = \frac{20}{1.802} = 11.09 \, A$$

7. Use the nodal analysis to find voltages V_a , V_b , V_c , V_d .



Solution



Apply KCL at node 1

$$i_m = i_x + i_y$$

$$-\frac{V_1}{10} = \frac{V_1 - V_2}{5} + \frac{V_1 - V_3}{20}$$

At node 2

$$10 + \frac{V_1 - V_2}{5} + \frac{V_3 - V_2}{20} = 0$$

At node 3

$$\frac{V_1 - V_3}{20} = \frac{V_3 - V_2}{10} + \frac{V_3}{5}$$

Solving,

$$V_1 = 45.45V, \quad V_2 = 72.73V \quad V_3 = 27.27V$$

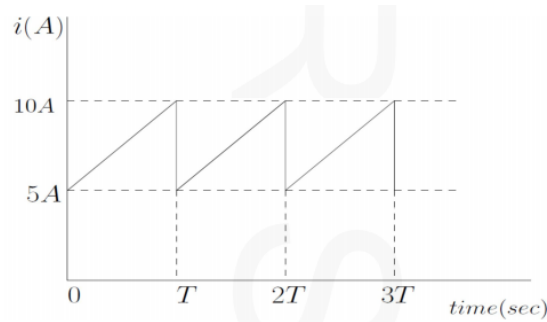
$$V_a = V_3 = 27.27V$$

$$V_b = -V_1 = -45.45V$$

$$V_c = V_1 - V_3 = 18.18V$$

$$V_d = V_3 - V_2 = -45.46V$$

8. Determine the average value and rms value of the current waveform shown in figure below.



Average value

$$I_{avg} = \frac{1}{T} \int_0^T v dt$$

$$\text{Average value} = \frac{\text{Area over Half cycle}}{\text{Base length}}$$

$$V_{avg} = \frac{\frac{1}{2} * b * h}{5} = \frac{\frac{1}{2} * (10 * T + 5 * T)}{T} = \frac{\frac{T}{2} * (10 + 5)}{T} = 7.5V$$

RMS Value

$$I_{rms} = \sqrt{\frac{1}{T} \int_0^T v^2 dt} = 7.64A$$

9. Define rms value and average value of a time varying wave form

Average value

- Average value of an alternating current is that steady current which transfers the same charge across any circuit, that transferred by the alternating current during the same time period.

$$V_{av} = \frac{\text{Area under the curve}}{\text{Time period}}$$

- Average value $I_{avg} = \frac{1}{T} \int_0^T i dt$

$$V_{avg} = \frac{1}{T} \int_0^T v dt$$

- RMS value(Root mean square)

The rms value of an alternating current is defined as the steady state dc current which when flowing through a given circuit for a given period of time produces the same heat as produced by the alternating current for the same circuit and for the same period of time.

The rms value is also called effective value of any waveform.

$$I = \sqrt{\frac{1}{T} \int_0^T i^2 dt} \quad \text{rms value of current}$$

$$V = \sqrt{\frac{1}{T} \int_0^T v^2 dt} \quad \text{rms value of voltage}$$

i, v are instantaneous value of current and voltage respectively.

10. A resistor of 50Ω , an inductor of 0.1H and a capacitor of $40\mu\text{F}$ are connected in series and the combination is connected across 220V , 50 Hz supply. Calculate (i) the circuit impedance (ii) resulting current (iii) power factor (iv) phase angle and (v) power consumed by the circuit.

$$R = 50\Omega$$

$$L = 0.1\text{ H}$$

$$C = 40\mu\text{F}$$

$$V = 220\text{V}$$

$$F = 50\text{Hz}$$

- (i) Circuit Impedance

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\sqrt{50^2 + \left(2\pi fL - \frac{1}{2\pi fC}\right)^2}$$

$$= \sqrt{50^2 + \left(2\pi * 50 * 0.1 - \frac{1}{2\pi * 40 * 10^{-6}}\right)^2}$$

$$= \sqrt{50^2 + (48.16)^2} = 69.42$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$= 50 - j48.16 = 69.42 \angle -43.92^\circ$$

- (ii) Current

$$\text{Current } I = \frac{|V|}{|Z|} = \frac{220}{69.42} = 3.169 \text{ A}$$

(iii) Power factor

$$\cos \phi = \frac{R}{Z} = \frac{50}{69.42} = 0.72 \text{ leading}$$

(iv) Phase angle

$$\phi = \cos^{-1}(0.72) = 43.92^\circ$$

(v) Power consumed

$$\text{Active power} = V I \cos \phi$$

$$220 * 3.169 * 0.72 = 500.76 \text{ W}$$

11. Three inductive coils, each with a resistance of 22Ω and an inductance of 0.05 H are connected in first in star and then in delta, to a 3 phase 415 V , 50 Hz supply. Calculate for both star and delta connections, (i) phase current and line current and (ii) total power absorbed.

$$R = 22 \Omega$$

$$L = 0.05 \text{ H}$$

- Star connection**

$$X_L = 2\pi f_L = 15.7$$

$$\text{Impedance per phase, } Z = \sqrt{22^2 + 15.7^2} = 27.02 \Omega$$

$$\text{Phase voltage, } V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{415}{\sqrt{3}} = 239.6 \text{ V}$$

$$\text{Phase current, } I_{ph} = \frac{V_{ph}}{Z} = \frac{239}{27.02} = 8.86 \text{ A}$$

$$\text{In star connected system, } I_L = I_{ph} = 8.86 \text{ A}$$

$$\text{Power factor, } \cos \phi = \frac{R}{Z} = \frac{22}{27.02} = 0.8142$$

$$\text{Power } P = \sqrt{3} V_L I_L \cos \phi = 15.5 \text{ KW}$$

$$P = \sqrt{3} * 415 * 8.86 * 0.8142 = 5.185 \text{ kW}$$

- Delta connection**

$$\text{Phase voltage, } V_{ph} = V_L = 415 \text{ V}$$

$$\text{Phase current, } I_{ph} = \frac{V_{ph}}{Z} = \frac{415}{27.02} = 15.35 \text{ A}$$

$$\text{In Delta connected system, } I_L = \sqrt{3} I_{ph}$$

$$= \sqrt{3} * 15.35 = 26.6 \text{ A}$$

$$\text{Power factor, } \cos \phi = \frac{R}{Z} = \frac{22}{27.02} = 0.8142$$

$$\text{Power } P = \sqrt{3} V_L I_L \cos \phi$$

$$P = \sqrt{3} * 415 * 26.6 * 0.8 = 15.5 \text{ kW}$$